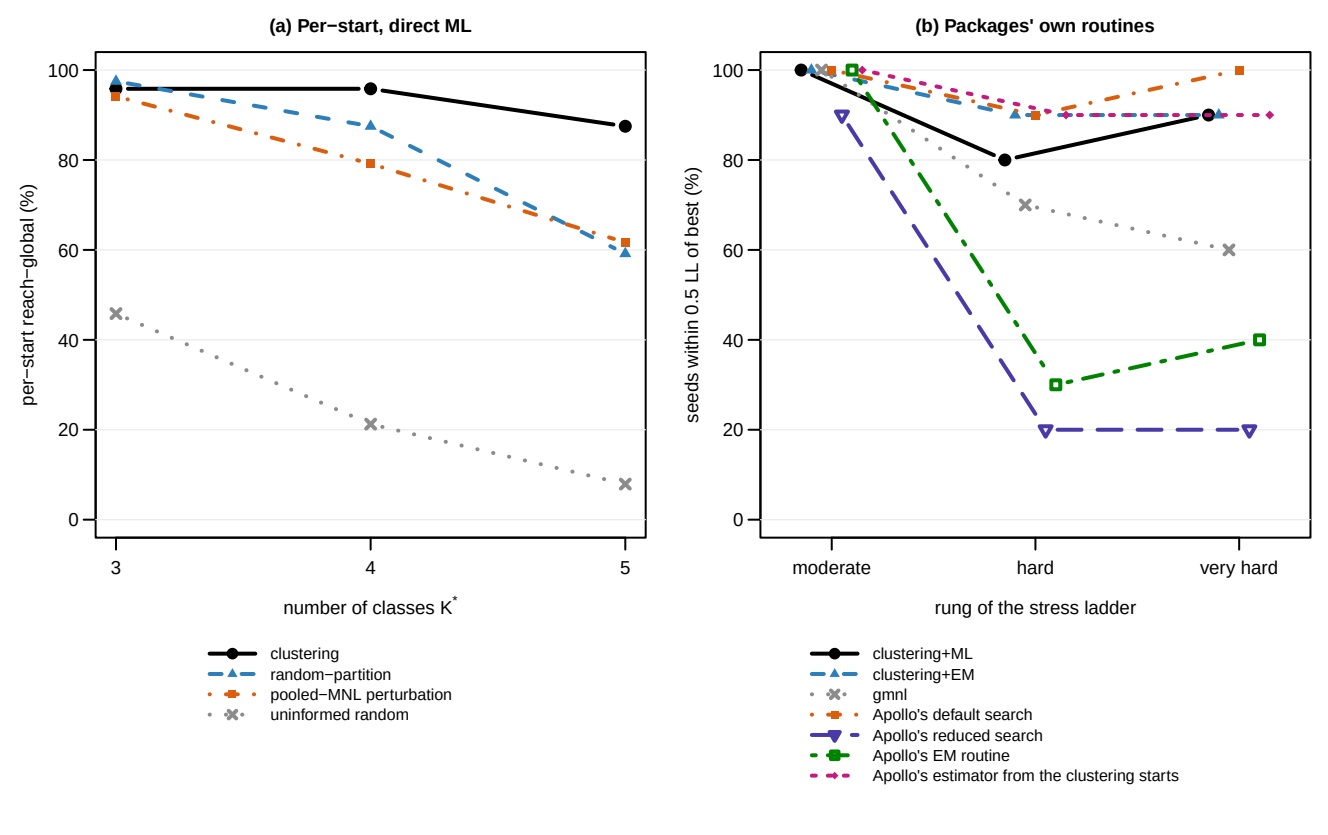
(a) Per−start, direct ML
per−start reach−global (%)
100
80
60
40
20
0
3
4
5
number of classes K*
clustering
random−partition
pooled−MNL perturbation
uninformed random
(b) Packages' own routines
seeds within 0.5 LL of best (%)
100
80
60
40
20
0
moderate
hard
very hard
rung of the stress ladder
clustering+ML
clustering+EM
gmnl
Apollo's default search
Apollo's reduced search
Apollo's EM routine
Apollo's estimator from the clustering starts